

Project Design Phase-II

Automated Network Request Management in ServiceNow

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Team ID: Individual / Solo Project

SCENARIO: An employee discovers, requests, gets approval for, and receives automated fulfillment of a network configuration change (e.g., VPN access, firewall rule) through the IT Service Catalog.

Phases	1. Awareness & Discovery <i>How does the employee know where to go?</i>	2. Portal Login <i>Entering the system</i>	3. Find Network Service <i>Locating the right form</i>	4. Submit Request <i>Filling out the details</i>	5. Approvals & Policy Check <i>Manager & InfoSec review</i>	6. Automated Fulfillment <i>Network is configured</i>	7. Verification & Close <i>Testing the connection</i>	8. CMDB Update <i>Logging the asset</i>	9. Audits & Expiration <i>Tracking lifecycle</i>	10. Optimization <i>Future improvements</i>
Steps What is the employee specifically doing?	IT Policy Review — Employee learns that network/port changes must be requested via the Service Catalog rather than emailing IT.	Log into ServiceNow — Employee opens the IT Service Portal and signs in via Single Sign-On (SSO).	Browse / Search Catalog — Employee navigates to "Network Services" or searches for "VPN" or "Firewall Rule".	Complete & Submit Form — Fills in Source IP, Destination IP, Port, and Business Justification. Submits request.	Workflow Routing — Request routes to Manager for sign-off. High-risk requests route to InfoSec. CMDB validates IPs.	IntegrationHub / Ansible — Upon approval, system triggers automation tool (e.g., Ansible) to push configuration to the firewall/router.	Test & Confirm — Employee receives completion notification, tests network access, and confirms ticket closure.	Topology Updated — The new network relationship/rule is automatically logged into the CMDB for accurate tracking.	Compliance Tracking — System tracks temporary access rules and triggers renewal/revocation workflows automatically.	Service Analytics — IT reviews dashboard to identify common requests to further standardize network catalog items.
Interactions What tools or systems are used?	Company Intranet, IT Policy page, HR Onboarding Checklist.	SSO Login screen, ServiceNow Employee Portal.	Service Catalog Homepage, Global Search Bar.	Network Request Catalog Item (variables, submit button).	ServiceNow Approval Engine, Manager Email/Teams, CMDB/SecOps engine.	Flow Designer, IntegrationHub, Ansible / Network Controller API.	Automated Email Notification, ServiceNow Ticket Portal.	ServiceNow CMDB module.	Flow Designer scheduled jobs, Expiration Notifications.	ServiceNow Reporting & Dashboards.
Goals & Motivations What is the employee trying to do?	"Help me understand the correct process to get network access."	"Help me get into the request system quickly."	"Help me find the exact network form I need without confusion."	"Help me provide the right technical details easily."	"Help me get quick sign-off without security blocking my work."	"Help me get connected immediately with zero IT delay."	"Help me confirm my system can now reach the server."	"Help IT track my access so I don't get audited later."	"Help me keep my access active until my project is done."	"Help make this process even faster next time."
Positive Moments What feels easy or reassuring?	Clear IT documentation points directly to the correct portal link.	SSO means no new passwords to remember.	Search bar immediately returns the correct "Firewall Port Request" item.	Form auto-populates employee details and validates IP formats.	Manager can approve instantly via a button in Microsoft Teams.	Zero-touch configuration means access is granted in minutes instead of days.	Prompt notification tells employee exactly what to test.	No manual spreadsheet tracking required by Network Admins.	Proactive renewal emails prevent unexpected access drops.	IT identifies trends to create pre-approved standard changes.
Negative Moments What is frustrating or slow?	Some users still try emailing the helpdesk and get rejected.	MFA/Authenticator app issues occasionally delay login.	If terminology differs (e.g., user types "unblock" instead of "firewall"), search might fail.	User might not know the exact Destination IP or Port required.	Request stalls if manager is on PTO; InfoSec rejects if justification is weak.	Automation fails if target device is offline or CMDB data is stale.	Employee forgets to close ticket; system auto-closes prematurely.	CMDB sync errors cause compliance reporting gaps.	Users ignore renewal emails and lose critical access mid-project.	Data is ignored; process remains rigid.
Areas of Opportunity How might we improve?	Set up auto-replies on IT email alias redirecting to the Catalog link.	Provide quick self-service password reset on login screen.	Add meta-tags to catalog items to catch common user synonyms.	Integrate CMDB dropdowns so users can select server names instead of typing IPs.	Configure auto-escalation/delegation in Flow Designer for absent managers.	Implement robust error-handling/retry logic in IntegrationHub.	Include a simple "Yes, it works" button in the closure email.	Run periodic Discovery scans to validate CMDB accuracy.	CC managers on expiration warnings to ensure business continuity.	Use AI/Predictive Intelligence to suggest network access based on user role.